Cpts 317. 9/14.

DFA — deterministic finite automaton. M

Intuition: M reads an (input) word letter by letter;
at the same time, on each letter read,
M moves one state to another;
M has an initial state;
M has an accepting/final state where
M chooses to stop;

mathematical def:

A deterministic finite automaton (DFA) is a tuple:

$$M = \langle \Sigma, Q, q_0, \delta, F \rangle$$

where: Σ : input alphabet ;

Q : set of states (Q is finite)

$q_0 \in Q$: initial state;

$F \subseteq Q$: set of accepting or final states;

$\delta : Q \times \Sigma \longrightarrow Q$, called transition function.

(Semantics of DFA).

Let $w = a_1 \cdots a_n$ be an input word, for some n.

M accepts w if there are states $q_0, q_1, \ldots, q_n$ in M s.t.

(1). q_0 is the initial state;

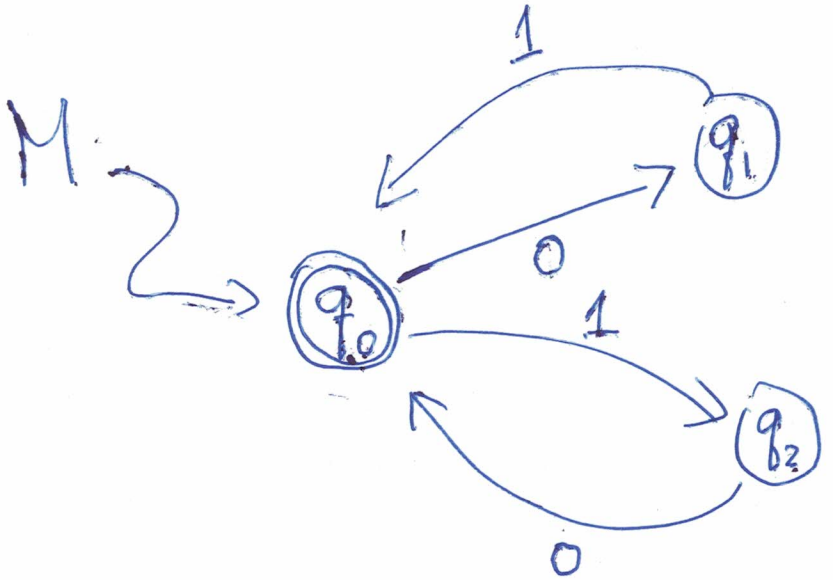
(2). $q_n \in F$

(3). for each $0 \le i \le n-1$, $\delta(q_i, a_{i+1}) = q_{i+1}$,

written as:

$$\rightarrow q_0 \xrightarrow{a_1} q_1 \xrightarrow{a_2} q_2 \cdots \xrightarrow{a_n} q_n.$$

In this case, we simply say $w \in L(M)$, where $L(M) = \{w : M \text{ accepts } w\}$, which is called



- - - - - - (*)

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this M is $\langle \Sigma, Q, q_0, \delta, F \rangle$ where

$\Sigma = \{0, 1\}$

$Q = \{q_0, q_1, q_2\}$, q_0 is the initial state;

$F = \{q_0\}$

δ is defined as: $\delta(q_0, 0) = q_1$; $\delta(q_0, 1) = q_2$

$\delta(q_1, 1) = q_0$; $\delta(q_2, 0) = q_0$.

transition table. $\Rightarrow$

the language accepted by DFA M.

Each DFA accepts exactly one language.

For the previous exaple in (*);

$01 \in L(M)$? yes, since

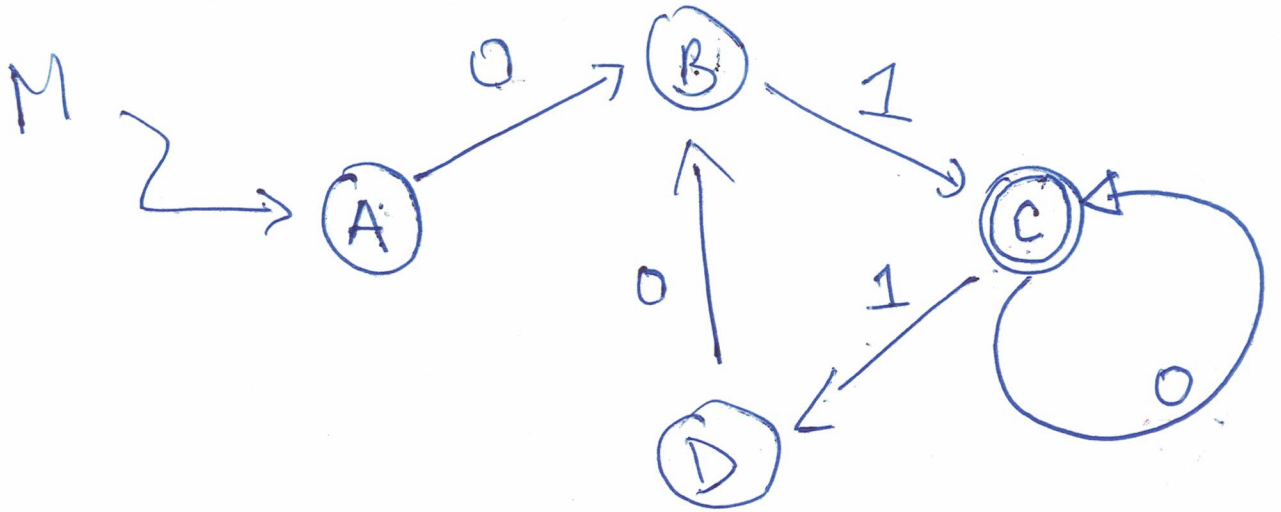
$$\rightarrow q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_0 \checkmark$$

$100 \in L(M)$? NO, since

$$\rightarrow q_0 \xrightarrow{1} q_2 \xrightarrow{0} q_0 \xrightarrow{0} q_1 \times$$

$$L(M) = (01 + 10)^*.$$

Second Example:



$0100101 \in L(M)$? yes, since

$$\to A \xrightarrow{0} B \xrightarrow{1} C \xrightarrow{0} C \xrightarrow{0} C \xrightarrow{1} D$$

$$\xrightarrow{0} B \xrightarrow{1} C \checkmark$$

$01001 \notin L(M)$, why?

$$\to A \xrightarrow{0} B \xrightarrow{1} C \xrightarrow{0} C \xrightarrow{0} C \xrightarrow{1} D \times$$

Big Challenge: $L(M) = 01(0^* + 101)^*$

$0111 \in L(M)$? NO, since

$$\rightarrow A \xrightarrow{0} B \xrightarrow{1} C \xrightarrow{1} D \longrightarrow \text{crash} \quad \times$$

Practical Intuition of FA:

state $=$ memory content.

k states $=$ memory of $\log_2 k$ bits.

2 states $=$ 1 bit of memory,

1024 states $=$ $\log_2 1024 = 10$ bits of memory;

DFA with 1024 states $\equiv$ a C-program with exactly 10 Boolean vars;

// $1024 = 2^{10}$

4 states $= \log_2 4 = 2$ bits of memory,

b_1	b_2	b_1	b_2	b_1	b_2	b_1	b_2
0	0	0	1	1	0	1	1
A		B		C		D	

Why do you come to WSU-CS?

Consider a program with 50 integer vars; (32 bit.)

declared an array of 50 integers;

$\|$

a DFA with $2^{50 \times 32}$ states

$\approx$

2^{1500}

$\approx$

① 3000 ?

② # of atoms in galaxy

③ # of bugs in your room.

how about a program with 500 int vars?

$\parallel$

DFA with $2^{500 \times 32}$ states.

# of atoms in the universe

Challenge of today:

how are you going to store this DFA as a graph in memory?

reason that you late gets 350.